

Synthesizing Static Data and Human Intent: A Comprehensive Survey of Offline and Preference-Based Reinforcement Learning for Robotics

1 Introduction

The deployment of autonomous robotic systems in unstructured physical environments is fundamentally constrained by two interdependent challenges: the prohibitive safety risks and sample costs associated with online exploration, and the inherent ambiguity in manually engineering scalar reward functions for multifaceted objectives. Traditional Reinforcement Learning (RL) relies on active trial-and-error interaction, a paradigm often infeasible for real-world robotics where failures entail physical damage or human injury [1]. Conversely, specifying precise reward functions for complex tasks, such as dexterous manipulation or mobile navigation, remains a cognitively demanding and error-prone process, frequently leading to reward hacking or misaligned behaviors [2; 3]. This survey posits that the convergence of Offline Reinforcement Learning (Offline RL) and Preference-Based Reinforcement Learning (PbRL) offers a rigorous pathway to overcome these dual bottlenecks. By leveraging static datasets to mitigate exploration risks and utilizing comparative human feedback to resolve reward ambiguity, this unified framework enables the learning of safe, high-performance policies without further environment interaction or explicit reward specification.

The historical evolution of this field reflects a trajectory from isolated paradigms toward integrated, data-efficient frameworks. Early approaches to learning from fixed data were dominated by Behavior Cloning (BC), which, while safe, suffers from compounding errors and an inability to exceed the performance of the demonstrator [4; 5]. Simultaneously, Inverse Reinforcement Learning (IRL) sought to infer rewards from demonstrations but often struggled with ill-posedness and scalability in high-dimensional spaces [6; 7]. The advent of deep Offline RL introduced mechanisms such as pessimistic value estimation and policy constraints to handle distributional shift, allowing agents to learn from suboptimal data while avoiding extrapolation errors [8; 9]. Parallel advancements in PbRL demonstrated that human preferences over trajectory segments could effectively substitute for dense reward signals, enabling the alignment of complex behaviors with nuanced human intent [2; 10]. Recent innovations have begun to bridge these domains, exploring methods that extract reward signals from ranked demonstrations or integrate preference feedback into offline optimization loops to achieve better-than-demonstrator performance [5; 11].

This survey provides a comprehensive taxonomy of the emerging landscape where offline data constraints intersect with preference-based alignment. We categorize existing literature based on three critical dimensions: data sources (e.g., expert demonstrations vs. heterogeneous replay buffers), feedback modalities (e.g., pairwise comparisons vs. listwise rankings), and algorithmic strategies (e.g., explicit reward modeling vs. direct policy optimization). A central theme of our analysis is the distinction between approaches that first infer a latent reward function before optimizing a policy, such as Bayesian reward extrapolation [12; 13], and those that bypass reward modeling entirely through direct contrastive objectives like Direct Preference Optimization (DPO) or Inverse Preference Learning (IPL) [14; 15]. Furthermore, we critically examine the theoretical challenges unique to this intersection, including the compounding effects of distributional shift and reward confusion, where limited data coverage can lead to the identification of spurious correlations rather than true intent [3; 16]. By synthesizing these diverse threads, this work establishes a unified foundation for developing robotic systems that are not only data-efficient and safe but also robustly aligned with complex human values, paving the way for scalable autonomy in dynamic real-world settings.

2 Theoretical Foundations and Problem Formulations

2.1 Mathematical Formalization of Offline Decision Processes with Partial Coverage

Offline reinforcement learning (RL) in robotics necessitates a rigorous departure from the idealized assumption of full dataset coverage, acknowledging that static collections of trajectories often exhibit significant distributional shift relative to optimal behaviors. Formally, we consider a Markov Decision Process (MDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma, \rho_0)$, where the learner accesses a dataset $\mathcal{D} = \{(s_i, a_i, r_i, s'_i)\}$ generated by a behavior policy μ . The fundamental challenge lies in the partial coverage of $\mathcal{D}$, quantified by the single-policy concentrability coefficient $C^* = \sup_{s,a} \frac{d^{\pi^*}(s,a)}{d^\mu(s,a)}$, which bounds the density ratio between the optimal policy’s occupancy measure and the data distribution [17]. When C^* is large or unbounded, standard off-policy evaluation fails due to the “curse of horizon,” where variance in importance sampling estimators scales exponentially with the time horizon [18]. Consequently, theoretical guarantees for sample-efficient learning require relaxing the all-policy concentrability assumption to single-policy concentrability, ensuring the data sufficiently covers the optimal policy rather than the entire state-action space [19].

To address the extrapolation errors inherent in partial coverage, the principle of pessimism has emerged as a critical theoretical mechanism. Pessimistic algorithms construct lower confidence bounds (LCB) on value functions, effectively penalizing uncertainty in out-of-distribution (OOD) regions to prevent the exploitation of model errors [20]. This approach flips the optimism bonus used in online exploration, yielding a minimax-optimal suboptimality bound that depends linearly on C^* rather than requiring uniform coverage [21]. Recent advancements refine this by introducing Bellman-consistent pessimism, which implements conservatism over the set of functions consistent with Bellman equations rather than via point-wise bonuses, thereby improving sample complexity by a factor of the feature dimension in linear MDPs [22]. Furthermore, variance-aware pessimistic value iteration leverages conditional variance information to reweight Bellman residuals, providing tighter instance-dependent bounds that adapt to the intrinsic difficulty of the task [23].

In the context of Inverse Reinforcement Learning (IRL) with offline data, the problem shifts from identifying a unique reward function to characterizing a feasible set of rewards consistent with observed demonstrations under partial coverage. The Clarity of Reward Extrapolation (CLARE) framework integrates conservatism directly into the learned reward function, balancing exploitation of expert data with exploration of estimated dynamics to mitigate covariate shift [24]. Similarly, distributionally robust formulations penalize reward estimates based on data-driven uncertainty, ensuring that the learned policy remains robust to perturbations in the nominal model even when the dataset only provides robust partial coverage [25].

Despite these theoretical strides, significant challenges remain regarding the interplay between function approximation realizability and coverage gaps. While pessimism resolves issues under single-policy concentrability, recent negative results indicate that without a gap assumption or stronger structural conditions like low-rank transitions, sample-efficient evaluation remains impossible even with trajectory data if only value function realizability is assumed [26]. Future directions must therefore focus on developing adaptive algorithms that can automatically tune the degree of pessimism based on local data density, potentially bridging the gap between imitation learning on expert data and conservative RL on heterogeneous datasets [27]. Ultimately, the mathematical formalization of offline decision processes hinges on precisely quantifying the trade-off between the breadth of data coverage and the depth of conservatism required to guarantee safe, near-optimal robotic control.

2.2 Probabilistic Modeling of Human Preferences and Latent Reward Structures

Building on the challenges of partial data coverage and the necessity of pessimism in standard offline RL, the transition to preference-based settings introduces the fundamental task of inferring latent reward structures from qualitative human feedback. The probabilistic modeling of these preferences serves as the theoretical cornerstone for bridging subjective human intent with objective robotic optimization. Central to this domain is the Bradley-Terry-Luce (BTL) model, which posits that the probability of preferring one trajectory segment σ_1 over another σ_2 is governed by a logistic function of their latent reward difference. While foundational works like [2] established the efficacy of BTL for Markovian rewards, recent theoretical scrutiny reveals significant limitations when human judgment depends on non-Markovian temporal dynamics or specific critical events rather than simple cumulative returns. [28] rigorously demonstrate that standard partial return models often lack identifiability, whereas regret-based preference models—measuring deviation from optimal decision-making—guarantee unique recovery of the underlying reward function given sufficient data. This distinction is critical for robotics, where safety constraints and goal completion often dictate preferences in a manner inconsistent with additive reward assumptions, thereby complicating the direct application of the single-policy concentrability bounds discussed previously.

To address the complexity of long-horizon dependencies and the limitations of Markovian assumptions, the field has evolved toward advanced architectures capable of capturing nuanced temporal structures. The [10] introduces a paradigm shift by utilizing self-attention mechanisms to model preferences as weighted sums of non-Markovian rewards, effectively attending to critical events within a trajectory that traditional recurrent or convolutional models might overlook. This architectural advancement allows for the disentanglement of multimodal state-action interactions, enabling the reward model to reflect the complex interplay driving human judgment in physical tasks. Furthermore, [13] and [29] highlight the synergistic potential of combining comparative feedback with demonstrations to ground the reward space. By leveraging explicit demonstrations, these approaches reduce the effective dimensionality of the inference problem and enhance sample efficiency, partially mitigating the coverage gaps inherent in purely preference-based offline datasets.

However, despite these architectural and data-integration advances, the recovery of a unique latent reward function from pairwise comparisons remains an ill-posed problem without strict identifiability conditions, echoing the realizability challenges noted in standard offline RL. [30] critiques the common approximation of substituting pairwise preferences with pointwise rewards, deriving a general objective Ψ PO that bypasses such assumptions to prevent policy misalignment. Theoretical analyses in [31] further establish that learning any target policy requires the offline dataset to satisfy specific single-policy concentrability coefficients, distinguishing the sample complexity of preference-based learning from standard reinforcement learning where rewards are directly observed. Moreover, the presence of hidden context—such as diverse annotator biases or unobserved environmental factors—can lead to implicit aggregation rules akin to Borda counts, potentially resulting in counter-intuitive policy behaviors as argued in [32]. Consequently, emerging frameworks like [33] advocate for latent variable formulations that infer user-specific parameters, allowing for pluralistic alignment rather than a single averaged reward function. These identifiability issues and latent variable complexities set the stage for the compounded theoretical difficulties arising from the intersection of offline data constraints and preference ambiguity, particularly regarding reward confusion and the need for double pessimism.

2.3 Theoretical Challenges in Distributional Shift and Reward Confusion

The intersection of offline data constraints and the inherent ambiguity of preference feedback introduces compounding theoretical challenges that distinguish offline preference-based reinforcement learning (PbRL) from both standard offline RL and online PbRL. A primary difficulty arises from reward confusion, where limited support in the offline dataset causes reward models to latch onto spurious correlations rather than the intended causal features of the task. [3] systematically demonstrates that optimizing misidentified rewards drives policies off the training distribution, resulting in high predicted rewards but catastrophic failure in true performance. This phenomenon is exacerbated when non-causal distractor features or partial observability are present, as the reward model lacks the active exploration necessary to disambiguate correlation from causation. Consequently, the transitivity of preferences, a foundational assumption in models like Bradley-Terry, may fail globally even if it holds locally within the data support, necessitating robust optimization strategies that account for these structural inconsistencies.

Theoretical analysis reveals that the sample complexity required to mitigate reward confusion is heavily instance-dependent, governed by the suboptimality gap and the specific coverage of the behavior policy relative to the optimal policy. Recent work on instance-optimal offline RL establishes intrinsic bounds that characterize the hardness of specific problem instances, showing that minimax rates often obscure the nuanced dependency on local data density [34]. In the preference-based setting, this complexity is further compounded by the information loss inherent in binary comparisons; unlike pointwise rewards, preferences provide only relative ranking information, which increases the variance of reward estimates in low-coverage regions. [35] theoretically proves that while preference-based RL can be reduced to robust reward-based RL, the reduction incurs costs related to the tolerance of reward errors, highlighting the statistical inefficiency when data coverage is partial. Furthermore, [30] argues that standard approximations substituting pairwise preferences with pointwise rewards introduce significant pitfalls, particularly when generalizing to out-of-distribution states sampled by the learned policy.

To address these coupled uncertainties in dynamics and preference models, recent theoretical frameworks advocate for “double pessimism.” This principle imposes conservatism not only on the value function estimates due to distributional shift but also on the reward model itself to prevent exploitation of preference noise. [25] proposes algorithms that achieve sample efficiency under robust partial coverage by penalizing both the mismatch between behavior and target policies and the perturbation of the nominal reward model. Similarly, [36] utilizes a dynamic discrete choice model to recover human rewards while employing pessimistic value iteration, providing guarantees that match classical offline RL bounds despite the additional layer of preference uncertainty. However, overly aggressive pessimism can lead to the “Matthew effect,” where popular actions are reinforced while diverse, potentially optimal behaviors are suppressed [37]. Future theoretical directions must therefore focus on adaptive pessimism mechanisms that dynamically balance the trade-off between mitigating reward confusion and preserving policy diversity, potentially leveraging causal inference to disentangle spurious correlations from true intent in high-dimensional robotic domains.

2.4 Comparative Analysis of Offline Preference Learning versus Online and Imitation Paradigms

Building on the theoretical challenges of reward confusion and double pessimism outlined previously, this subsection critically delineates the boundaries separating offline preference-based learning from its online counterparts and standard imitation learning paradigms. While all three approaches aim

to align robotic policies with human intent, they diverge fundamentally in their data coverage requirements, information efficiency, and safety guarantees. A primary theoretical distinction lies in the concentrability conditions necessary for convergence. Recent rigorous analysis demonstrates that offline contrastive methods, such as Direct Preference Optimization (DPO), typically necessitate a global coverage condition to guarantee convergence to the optimal policy [38]. In contrast, online Reinforcement Learning from Human Feedback (RLHF) can achieve optimality under weaker partial coverage assumptions, as active exploration allows the agent to “fill in the gaps” of the state-action space that static datasets miss [39; 40]. This separation elucidates the empirical performance gaps observed in data-scarce robotic domains, where offline methods may stagnate if the behavior policy fails to explore regions critical for optimal reward maximization, a limitation that active exploration in online settings can overcome.

Regarding information efficiency, preference signals offer a distinct advantage over expert demonstrations in resolving ambiguities within multimodal behavior distributions. Standard imitation learning, particularly Behavioral Cloning, is inherently limited by the performance of the demonstrator and suffers from covariate shift when the agent deviates from the training distribution [41; 4]. Conversely, preference-based frameworks can theoretically exceed the performance of the behavior policy by leveraging relative ranking information to infer a reward function that generalizes beyond the specific actions observed [5]. By integrating demonstrations to initialize a reward prior and refining it with preferences, frameworks like DemPref illustrate how comparative feedback can resolve reward function ambiguity more efficiently than demonstrations alone [13; 29]. Furthermore, methods such as D-REX validate that automatically generated rankings from noisy demonstrations can facilitate extrapolation beyond suboptimal behaviors without requiring explicit optimal labels [5].

Safety and extrapolation risks constitute the third axis of differentiation, directly addressing the trade-offs between the conservatism discussed earlier and the need for robustness. Offline preference learning inherits the safety constraints of the static dataset by strictly avoiding online exploration, thereby mitigating the risk of catastrophic failures associated with trial-and-error in physical environments [42]. However, as noted in the discussion on reward confusion, this safety comes at the cost of potential reward misidentification, where limited data support leads to reward models that exploit spurious correlations rather than true intent [43]. Online RLHF, while capable of correcting such misconceptions through active querying [44], introduces inherent risks where policy updates may lead the agent into unsafe states unknown during training [45]. Theoretical work suggests that offline methods can safely improve upon suboptimal policies if the dataset satisfies specific single-policy concentrability conditions and if pessimistic principles are applied to value estimation [27; 46]. Ultimately, the choice between these paradigms involves a trade-off between the sample efficiency and safety of offline learning versus the exploratory power and robustness of online interaction, suggesting that hybrid approaches leveraging the strengths of both may be required for scalable robotic deployment [47; 35].

3 Algorithmic Paradigms for Robust Offline Policy Optimization

3.1 Pessimistic Value Estimation and Uncertainty-Aware Critic Regularization

Pessimistic value estimation constitutes a foundational pillar in model-free offline reinforcement learning, addressing the critical challenge of extrapolation error induced by distributional shift. In the absence of active environment interaction, standard Bellman updates tend to overestimate the value of out-of-distribution (OOD) actions due to function approximation errors, leading to pol-

icy collapse. To mitigate this, contemporary algorithms enforce conservatism by modifying critic objectives to learn lower bounds on true value functions. A seminal approach in this domain is Conservative Q-Learning (CQL), which augments the standard Bellman error with a regularizer that penalizes high Q-values for actions not present in the dataset, thereby ensuring the learned Q-function serves as a theoretical lower bound on the policy’s true value [48]. This principle of pessimism has been rigorously justified theoretically; for instance, pessimistic value iteration algorithms have been shown to be minimax optimal under partial data coverage, effectively eliminating spurious correlations arising from insufficiently explored trajectories [20].

Beyond explicit regularization, uncertainty quantification plays a pivotal role in driving pessimistic updates. Ensemble methods have emerged as a practical mechanism to estimate epistemic uncertainty without requiring explicit density modeling of the behavior policy. By leveraging the disagreement among an ensemble of Q-networks, algorithms can dynamically penalize value estimates in regions of high uncertainty. However, recent analyses highlight that the independence of ensemble members is crucial; shared target values can paradoxically induce optimism, necessitating architectures like MSG that utilize completely separate networks to maintain valid lower confidence bounds [49]. Complementary approaches, such as Uncertainty Weighted Actor-Critic (UWAC), employ dropout-based uncertainty estimation to down-weight the contribution of OOD state-action pairs during training, enhancing stability especially in sparse human demonstration datasets [50]. Similarly, Pessimistic Bootstrapping for Offline RL (PBRL) utilizes bootstrapped Q-function disagreement to drive pessimistic updates, providing provable uncertainty quantification in linear MDPs [51].

An emerging trend involves shifting from point-wise pessimism to distributional risk sensitivity. Recognizing that average return maximization may be insufficient for safety-critical robotic tasks, methods like Conservative Offline Distributional Actor Critic (CODAC) extend pessimism to the entire return distribution. CODAC penalizes predicted quantiles for OOD actions, converging to a uniform lower bound on return quantiles and enabling the learning of risk-averse policies from risk-neutral data [52]. This aligns with Offline Risk-Averse Actor-Critic (O-RAAC), which optimizes for Conditional Value at Risk (CVaR), offering distributional robustness against natural shifts in the deployment environment [53]. Furthermore, the degree of conservatism need not be static; Confidence-Conditioned Value Functions allow policies to adapt their risk posture at evaluation time by conditioning the critic on a confidence parameter, bridging the gap between rigid pessimism and adaptive decision-making [54].

Despite these advancements, trade-offs remain between the tightness of the lower bound and the generalization capability of the value function. Overly aggressive pessimism can suppress beneficial generalization, leading to suboptimal policies that merely imitate the behavior policy. Mildly Conservative Q-learning (MCQ) addresses this by actively training OOD actions with proper pseudo-values, achieving a balance that prevents overestimation without hindering performance improvement [55]. Future research directions must focus on refining uncertainty estimation in high-dimensional continuous control and integrating these pessimistic principles with preference-based feedback loops, where reward ambiguity compounds the risks of distributional shift.

3.2 Policy Constraint Mechanisms and Behavior Regularization Strategies

While pessimistic value estimation implicitly discourages out-of-distribution (OOD) actions by learning conservative lower bounds on the value function, policy constraint mechanisms constitute a complementary foundational paradigm that addresses extrapolation error by explicitly re-

stricting the learned policy to the support of the behavior policy. Rather than modifying the critic to penalize uncertain values, behavior regularization imposes direct penalties on the divergence between the target policy π and the behavior policy π_β . The canonical formulation augments the standard actor-critic objective with a regularization term, typically expressed as $\mathcal{L}(\pi) = \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi} [56] - \alpha \cdot D(\pi(\cdot|s) || \pi_\beta(\cdot|s))$, where D denotes a divergence measure such as Kullback-Leibler (KL) or Maximum Mean Discrepancy (MMD). Early frameworks like [57] systematically evaluated these approaches, revealing that while complex architectural constraints often yield diminishing returns, simple behavior cloning hybrids can achieve robust performance across diverse continuous control tasks. Similarly, [58] argues for the necessity of explicitly modeling the behavior policy via score-based generative models to provide stable, high-fidelity constraints that prevent the selection of uncertain actions more effectively than indirect value penalization.

However, global regularization strategies often suffer from over-conservatism, uniformly penalizing deviations even in state regions where the dataset provides dense coverage and the behavior policy is suboptimal. To mitigate this, recent advancements have shifted towards adaptive, state-dependent constraint relaxation. Methods such as [59] highlight that real-world data often exhibits heteroskedasticity, where action variability changes non-uniformly across the state space; consequently, fixed constraints fail to exploit high-quality regions. Addressing this, [60] proposes relaxing the policy constraint dynamically, tipping the balance from imitation to pure reinforcement learning as training progresses or as confidence in specific state-action pairs increases. Furthermore, [61] introduces a softer constraint mechanism that regularizes the policy towards the nearest valid state-action pair in the dataset rather than enforcing strict distribution matching, thereby allowing the agent to synthesize novel actions that remain within the feasible support manifold. This concept of support matching is further refined in [62], which utilizes variational autoencoders to explicitly model the density-based support set, ensuring that the learned policy avoids OOD regions without being tethered to the specific modes of a potentially multimodal behavior policy.

The choice of divergence measure also critically influences algorithmic stability and expressiveness, particularly when dealing with the complex action distributions common in robotics. While KL divergence is standard, it can be restrictive for multimodal behaviors. [63] demonstrates that representing policies as conditional diffusion models allows for high-fidelity behavior cloning while enabling policy improvement through guided sampling, effectively capturing complex multimodal distributions that Gaussian policies cannot. Additionally, [64] explores Fisher divergence as an alternative regularizer, linking behavior regularization to score matching and enabling more stable gradient flows in high-dimensional action spaces. Despite these advances in explicit policy constraints, a fundamental trade-off remains between staying close to the data to ensure safety and deviating sufficiently to surpass suboptimal demonstrators. Future research must therefore focus on adaptive mechanisms that can automatically calibrate constraint strength based on local data density and uncertainty estimates, potentially integrating causality-aware filters to distinguish between spurious correlations and genuine behavioral constraints. These explicit regularization techniques stand in contrast to the model-based approaches discussed in the following section, which seek to overcome the limitations of static datasets by leveraging learned dynamics to generate synthetic trajectories beyond the strict support of the behavior policy.

3.3 Model-Based Offline Learning with Dynamics Uncertainty Quantification

Model-based offline reinforcement learning (RL) leverages learned dynamics models to generate synthetic trajectories, thereby augmenting scarce static datasets and enabling planning beyond the strict support of the behavior policy. However, the fundamental challenge in this paradigm is

preventing the policy from exploiting epistemic uncertainties in the learned model, a phenomenon known as model exploitation. To address this, modern algorithms rigorously quantify dynamics uncertainty and incorporate pessimism directly into the planning objective. A foundational approach involves training ensembles of dynamics models to estimate epistemic uncertainty through disagreement among members. The Model-based Offline Policy Optimization (MOPO) algorithm exemplifies this by penalizing the reward function with the estimated uncertainty of the dynamics prediction, theoretically maximizing a lower bound on the true return under the real Markov Decision Process [65]. Similarly, MOREL constructs a pessimistic MDP where the transition dynamics are modified to ensure that performance in the surrogate environment lower-bounds performance in the true environment, effectively mitigating model exploitation while permitting generalization to unseen states [66].

While ensemble methods provide robust uncertainty estimates, they can be computationally demanding and sensitive to hyperparameters. Alternative strategies seek to induce conservatism without explicit uncertainty quantification. For instance, COMBO (Conservative Offline Model-Based Policy Optimization) regularizes the value function on out-of-support state-action tuples generated via model rollouts, achieving conservative value estimation without requiring explicit uncertainty bounds or complex ensemble architectures [67]. This approach simplifies implementation while maintaining theoretical guarantees on policy improvement. Further refining the interaction between model uncertainty and value estimation, recent works have explored pessimism-modulated dynamics beliefs, where policy optimization is performed via biased sampling from a belief distribution over dynamics, naturally inducing a policy-dependent reweighting factor that balances model utilization and risk avoidance [68]. Additionally, the “edge-of-reach” problem highlights that even with accurate dynamics, standard Bellman backups can fail at the boundary of the dataset; methods like Reach-Aware Value Learning (RAVL) address this by explicitly constraining value updates to reachable regions, ensuring stability in model-based imagination [69].

In high-dimensional visual domains, learning dynamics in pixel space is often intractable. Consequently, latent-space modeling has emerged as a critical direction, where dynamics are learned in a compressed representation to facilitate robust planning. Frameworks such as MOTO demonstrate that pre-training world models offline and fine-tuning them with uncertainty-aware planning allows for efficient adaptation to novel tasks from pixels, bridging the gap between offline pre-training and online flexibility [70]. Furthermore, hybrid approaches like H2O integrate imperfect simulators with offline data, using dynamics-aware evaluation to penalize Q-values where simulation-to-real gaps are large, effectively leveraging synthetic data while respecting real-world constraints [71]. Despite these advances, challenges remain in balancing the trade-off between the generalization benefits of model rollouts and the risk of distributional shift. Future research directions point towards more efficient uncertainty quantification mechanisms, such as single well-calibrated autoregressive models that may outperform large ensembles [72], and the integration of causal structures to ensure counterfactual generalization in multi-object settings [73]. ultimate success in robotic deployment will depend on algorithms that can dynamically modulate their reliance on learned models based on real-time confidence estimates.

3.4 Sequence Modeling and Generative Architectures for Trajectory Stitching

Bridging the gap between the dynamics-aware planning of model-based methods and the robustness challenges of heterogeneous data, the paradigm of offline reinforcement learning (RL) has undergone a transformative shift toward conditional sequence generation. This approach reframes policy optimization as a problem of modeling trajectory distributions, leveraging high-capacity

architectures—particularly Transformers and diffusion models—to address the critical challenge of “trajectory stitching.” Unlike value-based methods that often struggle with extrapolation errors in sparse data regimes, or model-based approaches constrained by epistemic uncertainty, sequence modeling treats the agent’s history, desired return, and future actions as a unified token stream to synthesize optimal behaviors by composing high-reward sub-segments from disjoint, suboptimal demonstrations. The seminal Decision Transformer framework established that conditioning action generation on returns-to-go via self-attention mechanisms enables effective long-horizon credit assignment and stitching across diverse tasks [74]. However, standard return-conditioned supervised learning implicitly assumes that high-return trajectories exist within the dataset, a condition rarely met in realistic robotic settings where data is predominantly suboptimal [75].

To mitigate this limitation and align generative capabilities with optimality principles, recent advancements integrate dynamic programming directly into the sequence modeling objective. The Q-learning Decision Transformer (QDT) and Q-value regularized Transformer (QT) address the inconsistency between sampled returns and optimal value estimates by utilizing Bellman backups to relabel training targets, thereby explicitly enforcing stitching capabilities that pure supervised approaches lack [76; 77]. Similarly, the Reinformer incorporates a max-return objective to guide the model toward predicting actions associated with the maximum feasible future return within the data distribution, enhancing performance in stitching-critical benchmarks [78]. Further refining temporal abstraction, the Waypoint Transformer introduces intermediate targets to seamlessly connect fragmented trajectory segments, demonstrating significant gains in complex navigation tasks where direct end-to-end modeling fails [79]. Beyond Transformers, alternative architectures like MambaDM exploit state-space models to efficiently capture both local Markovian dynamics and global temporal dependencies, offering a computationally scalable alternative for multi-scale sequence modeling [80].

Parallel to autoregressive models, diffusion probabilistic models have emerged as a powerful class for representing the multimodal action distributions inherent in heterogeneous robotic datasets, complementing the uncertainty quantification strategies discussed in model-based contexts. By formulating policy learning as a denoising process, Diffusion-QL couples behavior cloning with policy improvement, effectively constraining actions to the data support while optimizing for high values [63]. This generative capability is particularly potent for trajectory stitching; DiffStitch employs diffusion-based data augmentation to systematically generate synthetic transitions that bridge low-reward and high-reward regions, effectively constructing globally optimal paths from fragmented experiences [81]. In goal-conditioned settings, the SSD method leverages conditional diffusion to stitch sub-trajectories based on value estimates, overcoming the sparsity of reward signals in long-horizon tasks [82]. Furthermore, frameworks like AlignDiff extend these generative capabilities to preference alignment, guiding diffusion planners with attribute strength models to customize behaviors without retraining [83].

Despite these advances, challenges remain regarding optimism bias in stochastic environments, where joint modeling of states and actions can lead to unsafe predictions, echoing the distributional shift concerns raised in the context of imperfect data [84]. Future directions point toward hybrid architectures that combine the planning robustness of value functions with the expressiveness of generative models, such as using diffusion world models for conservative value regularization [85] or employing bootstrapped transformers to expand data coverage via self-generation [86]. The integration of these sequence modeling techniques with preference-based feedback loops promises to further enhance the adaptability and safety of offline robotic agents, setting the stage for addressing the specific complexities of data contamination and robust optimization discussed subsequently [10;

87].

3.5 Robust Optimization under Data Contamination and Distributional Shift

Real-world robotic datasets are rarely pristine; they often comprise heterogeneous sources ranging from expert demonstrations to suboptimal, noisy, or even adversarial trajectories. This subsection addresses the critical challenge of robust optimization under such data contamination and distributional shift, where standard offline reinforcement learning (RL) algorithms may fail due to extrapolation errors or the assimilation of spurious correlations. The core objective is to develop algorithmic paradigms that can discern signal from noise without prior knowledge of data quality, ensuring policy reliability in safety-critical domains.

A primary strategy for handling mixed-quality data involves reweighting or filtering trajectories based on inferred optimality. [88] proposes Return-based Data Rebalance (ReD), a simple yet effective method that constructs a better behavior policy by resampling transitions according to episodic returns. By aligning the support rather than the density of the distribution, ReD mitigates the negative impact of inferior actions while avoiding the strict constraints that limit performance upper bounds. Similarly, [89] introduces a quality-weighted loss regularization within a skill-based framework, decomposing tasks into subtasks consistent with high-quality skills to robustly handle heterogeneous datasets. For scenarios involving explicit corruption or adversarial attacks, [90] presents a certification framework that guarantees per-state action stability and cumulative reward bounds, utilizing temporal aggregation protocols to tolerate a specific fraction of poisoned trajectories.

Beyond reweighting, pessimistic principles serve as a foundational mechanism for robustness against distributional shift. The concept of “double pessimism,” articulated in [25], addresses shifts arising from both behavior policy mismatch and environmental perturbations. By combining pessimistic model estimation with pessimistic policy optimization, algorithms like P^2MPO achieve sample efficiency even under robust partial coverage. This aligns with [91], which employs a data-driven penalty term within a minimax formulation to learn policies that perform well under worst-case dynamics deviations specified by Kullback-Leibler divergence. Furthermore, [92] tackles imperfect reward signals through Reward Gap Minimization (RGM), a bi-level optimization that corrects reward biases by matching visitation distributions to expert data while solving a pessimistic RL problem.

Emerging trends also leverage representation learning and semi-supervised techniques to enhance robustness. [93] utilizes contrastive learning to derive task representations invariant to behavior policy shifts, crucial for meta-learning across diverse data sources. In semi-supervised settings where reward labels are scarce or noisy, [16] demonstrates that assigning zero rewards to unlabeled data can effectively trade off reward bias and distributional shift, often outperforming complex reward modeling. Additionally, [94] addresses the “type mismatch” in heterogeneous video data by pre-training value functions, enabling robust policy learning from observation-only internet-scale datasets.

Despite these advances, significant challenges remain in balancing conservatism with adaptability. Overly pessimistic approaches may stifle improvement in well-covered regions, while aggressive filtering risks discarding valuable exploratory data. Future directions should focus on adaptive robustness mechanisms that dynamically adjust confidence bounds based on local data density and uncertainty estimates, as well as unified frameworks that jointly optimize for robustness against both dynamic shifts and preference corruption in human-in-the-loop systems.

4 Preference Inference, Reward Modeling, and Direct Alignment

4.1 Reward Modeling Architectures and Latent Objective Inference

The architectural evolution of reward modeling in robotics has transitioned from restrictive linear feature combinations to expressive deep neural networks capable of decoding complex, non-Markovian human intent. Traditional approaches often assumed that human preferences were derived solely from the cumulative sum of Markovian rewards; however, empirical evidence suggests that human judgment is frequently driven by specific critical events, temporal patterns, and regret-based deviations rather than simple additive returns [28]. To address this, modern architectures like the Preference Transformer leverage self-attention mechanisms to model preferences as weighted sums of trajectory segments, effectively capturing long-horizon dependencies and attending to salient events that traditional recurrent or convolutional models miss [10]. This shift enables the inference of latent objectives that are robust to the distributional shifts inherent in offline datasets, where the behavior policy may not cover the optimal trajectory space.

A critical challenge in inferring these latent objectives is the inherent ambiguity and noise in human feedback, which can lead to reward misidentification and causal confusion if the model overfits to spurious correlations [3]. Probabilistic frameworks have emerged as a rigorous solution, modeling rewards as distributions rather than point estimates to explicitly quantify uncertainty arising from conflicting annotators or sparse data. For instance, Bayesian Reward Extrapolation (Bayesian REX) utilizes self-supervised pre-training to learn low-dimensional feature encodings, enabling efficient Bayesian inference over reward functions that scales to high-dimensional control problems while providing high-confidence performance bounds [12]. Similarly, active preference-based Gaussian Process regression allows for the learning of expressive, non-parametric reward functions without assuming rigid structural constraints, thereby balancing flexibility with data efficiency [44]. By incorporating uncertainty quantification, these methods facilitate safer policy optimization, preventing agents from exploiting regions of the state space where the reward model is uncertain—a phenomenon known as reward hacking.

Furthermore, the reality of pluralistic human values necessitates architectures capable of handling multimodal preference distributions rather than assuming a single universal reward function. Standard aggregation methods, such as those implicit in the Bradley-Terry-Luce model, often fail to capture heterogeneous objectives, potentially leading to “preference collapse” where minority views are disregarded [95]. To mitigate this, latent variable models like the Pluralistic Alignment (PAL) framework employ mixture modeling to infer user-specific latent variables, allowing for the simultaneous learning of multiple reward functions conditioned on individual intent without requiring separate models for each profile [96]. Complementing this, Distributional Preference Learning (DPL) estimates a distribution of possible score values to account for hidden contexts, such as varied annotator backgrounds or cognitive biases, thereby reducing vulnerabilities to adversarial exploitation [32].

Despite these advancements, robust learning from noisy and inconsistent feedback remains an open frontier. Recent methodologies incorporate sample selection discriminators to filter denoised preferences dynamically and utilize warm-start strategies to bridge the gap between pre-training and online refinement, significantly enhancing robustness in limited-feedback regimes [97]. Additionally, the integration of equal preference signals and the development of regularized objectives that prevent over-optimization are crucial for maintaining alignment integrity [98; 99]. Future directions lie in synthesizing these architectural innovations with causal inference techniques to disentangle true intent from environmental artifacts, ensuring that learned reward models generalize reliably across

diverse robotic embodiments and task domains.

4.2 Direct Preference Optimization and Reward-Free Paradigms

While the previous subsection detailed advancements in explicit reward modeling, the inherent instability of separating reward learning from policy optimization has spurred a paradigm shift toward Direct Preference Optimization (DPO) and related reward-free paradigms. By reparameterizing the optimal policy directly in terms of human preferences and a reference policy, these methods decouple alignment from the explicit construction of a scalar reward landscape. This transformation recasts the complex reinforcement learning problem as a supervised classification task, thereby eliminating the computational overhead and convergence issues associated with training separate reward models and performing online sampling [14]. Theoretical analysis confirms that DPO implicitly solves a regularized reinforcement learning objective, optimizing the likelihood of preferred trajectories while maintaining proximity to a behavior policy via a Kullback-Leibler divergence constraint [30]. Crucially, this formulation mitigates the “reward hacking” phenomena discussed earlier, where policies exploit errors in misspecified reward functions, offering a more robust path to alignment in data-constrained robotic domains.

Building on this foundation, contrastive approaches have been extended beyond the standard Bradley-Terry framework to address specific limitations in robotic control. For instance, Contrastive Preference Learning (CPL) derives objectives based on regret rather than cumulative return, positing that human preferences inherently reflect deviation from optimality rather than absolute utility; this enables fully off-policy learning without the need for reward bootstrapping [100]. Similarly, Inverse Preference Learning (IPL) leverages the interchangeability of Q -functions and reward functions for a fixed policy to eliminate the reward model entirely, achieving competitive performance in continuous control tasks with significantly fewer parameters [15]. In offline settings, algorithms like Offline Preference-guided Policy Optimization (OPPO) integrate hindsight information matching to jointly model trajectories and preferences, effectively stitching suboptimal segments without an intermediate reward bottleneck [101].

To further enhance robustness against the complexities of human judgment, such as intransitivity and stochasticity, recent works have adopted game-theoretic formulations. Self-Play Preference Optimization (SPPO) and the Minimaxalist approach frame alignment as a two-player zero-sum game, seeking the Nash equilibrium (or Minimax Winner) rather than the von Neumann winner assumed by standard logistic models [102; 103]. These methods provably converge even when preferences violate the axioms of rational choice, a critical feature for aggregating diverse human feedback in real-world robotics. Furthermore, generalizations like f -DPO introduce diverse divergence constraints beyond reverse KL, enabling a tunable balance between alignment strictness and generation diversity [104].

Despite their theoretical elegance, direct methods face distinct challenges regarding distributional shift and data quality. The implicit reward model induced by DPO has been shown to generalize less effectively than explicit models under significant distribution shift, potentially limiting performance in out-of-distribution robotic scenarios [105]. Additionally, naive application can lead to “preference collapse” or excessive verbosity due to algorithmic biases in length regularization [106; 95]. Consequently, future directions must focus on robust variants that incorporate uncertainty estimation, such as Value-Incentivized Preference Optimization (VPO), which unifies online and offline settings by regularizing with value functions to enforce optimism or pessimism [107], and methods that dynamically filter noisy preferences to prevent degradation from erroneous labels [108;

109]. While these strategies strengthen the direct optimization framework, they do not fully resolve the prohibitive cost of acquiring high-quality human feedback, a bottleneck that necessitates the shift toward active preference elicitation strategies discussed in the following section.

4.3 Sample-Efficient Feedback Elicitation and Active Querying

The prohibitive cost of acquiring high-quality human feedback necessitates a shift from passive data collection to active preference elicitation, where agents strategically query annotators to maximize information gain per interaction. In the context of offline and preference-based robotic learning, this challenge is formalized as a contextual dueling bandit problem, wherein the objective is to minimize regret while adhering to strict query budgets. Theoretical advancements have established that naive uniform sampling of trajectory pairs leads to suboptimal convergence, whereas active querying strategies can achieve sample complexity bounds that scale favorably with the feature dimension and sub-optimality gap [110; 111]. Specifically, algorithms leveraging uncertainty estimates in the reward model or policy space can identify query pairs that lie near the decision boundary, thereby resolving ambiguity in the latent reward function more efficiently than random exploration [44].

A critical innovation in this domain is the synthesis of informative queries using learned dynamics models, which allows agents to generate hypothetical trajectory pairs in simulation without risking physical safety or incurring real-world execution costs. By leveraging ensemble disagreement in dynamics or reward predictions, methods such as those proposed in [112; 113] incentivize the agent to explore regions of high epistemic uncertainty, effectively turning the reward model itself into an exploration bonus. This approach is particularly potent when combined with offline datasets; rather than discarding unlabeled experience, frameworks like [114] utilize semi-supervised techniques to infer pseudo-labels for vast amounts of unannotated data, reserving human queries only for instances where model confidence is lowest. Furthermore, the integration of offline data enables pool-based active learning, where the algorithm selects the most discriminative pairs from a static buffer of diverse behaviors, significantly reducing the need for online rollouts during the alignment phase [43].

Beyond simple pairwise comparisons, emerging strategies exploit the structure of human feedback to further enhance efficiency. For instance, incorporating textual rationales alongside binary preferences allows large language models to reason about state-action relevance, effectively expanding the information density of each query [115]. Additionally, the concept of “co-active” learning demonstrates that users can provide incremental improvements to proposed trajectories rather than absolute optimal demonstrations, a feedback modality that is often more intuitive and less cognitively demanding for humans while still providing strong theoretical regret guarantees [116; 117]. However, these active methods face challenges when human preferences are noisy, intransitive, or heterogeneous. Recent work addresses this by developing robust aggregation mechanisms that estimate user reliability and detect minority viewpoints within crowdsourced feedback, ensuring that active querying does not overfit to biased or erroneous annotators [118].

Looking forward, the frontier of sample-efficient elicitation lies in generalized acquisition functions that optimize for behavioral equivalence rather than precise reward parameter identification. By focusing queries on distinctions that materially alter the optimal policy, rather than irrelevant nuances in the reward landscape, future systems can achieve alignment with minimal human burden. This paradigm shift, supported by theoretical reductions of preference learning to robust reward-based RL [35], promises to make high-fidelity alignment feasible for complex, high-dimensional robotic tasks where exhaustive labeling is impossible.

4.4 Integrated Offline-Preference Frameworks and Hybrid Pipelines

While active elicitation strategies significantly reduce the burden of human feedback, the ultimate scalability of preference-based robotics relies on effectively integrating these sparse signals with vast static offline datasets. This synthesis addresses the dual challenges of distributional shift and reward ambiguity by creating hybrid pipelines that leverage historical data for robust initialization while utilizing human preferences for fine-grained alignment. A primary strategy in this domain involves transforming pairwise preferences into scalar pseudo-rewards to facilitate the application of established offline reinforcement learning (RL) algorithms. For instance, Optimal Transport Reward labeling (OTR) computes an optimal alignment between unlabeled trajectories and expert demonstrations to derive similarity-based rewards, allowing standard offline RL methods to match ground-truth reward performance with minimal supervision [119]. Similarly, frameworks have been proposed to craft preference queries via pool-based active learning from offline data, learning a distribution over reward functions that guides policy optimization without requiring physical rollouts [43]. These approaches effectively decouple the learning of dynamics from the inference of objectives, mitigating the information bottlenecks associated with separate reward modeling stages [101].

Moving beyond two-stage pipelines, recent advancements advocate for the joint optimization of rewards and policies under strict distributional constraints. The Offline Preference-guided Policy Optimization (OPPO) paradigm exemplifies this by modeling trajectories and preferences in a unified one-step process, utilizing an offline hindsight information matching objective to eliminate the need for explicit reward function learning [101]. This direct alignment is further supported by Inverse Preference Learning (IPL), which posits that for a fixed policy, the Q-function encodes all necessary reward information, thereby enabling parameter-efficient learning directly from offline preference data [15]. However, the efficacy of such purely offline methods is theoretically bounded by data coverage; rigorous analysis indicates that offline contrastive methods require global coverage to converge to optimal policies, whereas online methods can succeed with partial coverage [38]. To bridge this gap, hybrid strategies like Dataset Reset Policy Optimization (DR-PO) integrate offline preference datasets into online training by resetting the policy optimizer to informative states found in the static data, theoretically guaranteeing performance at least as good as any covered policy [120].

Furthermore, adapting pre-trained policies to specific user preferences necessitates efficient fine-tuning mechanisms that account for pluralistic human intent. The Pluralistic Alignment Framework (PAL) addresses this by employing mixture modeling to capture diverse population preferences within a shared latent space, enabling few-shot generalization to unseen users [96]. In multi-objective settings, policy-regularized offline RL extends single-objective constraints to handle preference-inconsistent demonstrations by filtering data based on approximated behavior preferences [121]. Additionally, leveraging unlabeled data through semi-supervised frameworks like SURF significantly improves feedback efficiency by inferring pseudo-labels based on predictor confidence, thus maximizing the utility of large-scale static datasets [114]. Despite these advances, challenges remain in balancing the conservatism required for offline safety with the exploration needed for preference refinement. Future directions likely involve developing adaptive uncertainty quantification methods that dynamically relax constraints in well-covered regions while maintaining robustness against spurious correlations in sparse data regimes, ensuring scalable and safe deployment of aligned robotic systems.

5 Integrated Frameworks for Offline Preference-Based Robotic Control

5.1 Hybrid Pre-training and Warm-Starting Architectures

Hybrid pre-training and warm-starting architectures address the fundamental sample inefficiency and cold-start limitations inherent in pure online preference-based reinforcement learning (PbRL) by leveraging large-scale static datasets to initialize policy, value, or reward components prior to alignment. The core premise involves decoupling the acquisition of broadly transferable representations from the refinement of task-specific intent, thereby reducing the query complexity required for human feedback. A primary strategy involves value-function pre-training on unlabeled offline trajectories, where temporal-difference learning or successor representations are employed to learn robust state embeddings and value priors from vast, heterogeneous datasets, including internet-scale videos [122]. These priors serve as a calibrated initialization, ensuring that subsequent fine-tuning with sparse human preference labels aligns with specific objectives without requiring dense action annotations during the pre-training phase [123]. This approach effectively mitigates the risk of distributional shift by grounding the agent in the support of the offline data before introducing the ambiguity of relative human feedback.

Complementing value-based initialization, skill-driven architectures extract diverse libraries of reusable primitives from offline data using unsupervised or self-supervised objectives [124]. By operating in a lower-dimensional skill space, these methods enable preference feedback to select, compose, or refine high-level behaviors rather than optimizing low-level actions directly, significantly accelerating convergence [89]. This hierarchical decomposition allows the agent to leverage “trajectory stitching” capabilities, composing optimal sub-segments from suboptimal demonstrations to satisfy complex preference constraints [125]. Furthermore, recent advances in representation learning utilize implicit preference signals within offline data—such as the natural alignment between visual trajectories and language instructions—to pre-train embodied multimodal representations [126]. This creates a foundation where the reward model itself can be warm-started using synthetic or implicit preferences derived from heuristic returns or trajectory diversity, stabilizing the initial reward landscape against noisy human feedback [44].

The integration of these paradigms often follows a sequential pipeline: unsupervised pre-training establishes a broad behavioral prior, followed by supervised fine-tuning on demonstrations, and finally, direct preference optimization for alignment [14]. However, emerging frameworks advocate for joint initialization where reward models are pre-trained on pseudo-labels generated from offline data before engaging in active querying [43]. This hybridization reduces the sensitivity of the reward model to early-stage noise, a critical factor given that reward misspecification can propagate catastrophically during policy updates [109]. Moreover, meta-learning approaches have been adapted to this setting, pre-training preference models on prior task distributions to enable rapid adaptation to new user intents with only a handful of queries [127]. Despite these advances, challenges remain in balancing the fidelity to offline data with the flexibility to exceed behavior policy performance; overly conservative warm-starts may inhibit exploration of preferred behaviors lying outside the initial data support. Future directions likely involve dynamic regularization schedules that relax offline constraints as preference confidence increases, and the development of foundation policies capable of zero-shot adaptation to diverse preference profiles through directional control in latent reward spaces [128].

5.2 Joint Optimization of Rewards and Policies from Static Data

Building upon the adaptive mechanisms and unified objectives introduced in hybrid warm-starting strategies, the joint optimization of rewards and policies from static data represents a critical synthesis in offline preference-based robotic control. This paradigm directly addresses the dual challenges of distributional shift and reward identifiability without requiring active environment interaction, effectively blurring the distinction between initialization and alignment. Unlike the sequential pipelines discussed previously, which first learn a reward model and subsequently optimize a policy, joint approaches formulate a unified objective to simultaneously infer latent reward structures and derive optimal behaviors, thereby mitigating the error propagation inherent in two-stage methods. A foundational strategy involves transforming pairwise preference feedback embedded within offline logs into scalar reward estimates via optimal labeling techniques, enabling the direct application of conservative offline reinforcement learning algorithms. For instance, frameworks leveraging optimal transport principles can align unlabeled trajectories with expert demonstrations to generate pseudo-rewards, effectively bridging the gap between preference data and standard offline RL solvers like Conservative Q-Learning [119]. This approach allows for the theoretical bounding of extrapolation errors by enforcing pessimism on out-of-distribution actions, ensuring that the learned policy remains within the support of the behavior policy [129].

Beyond explicit reward labeling, emerging paradigms utilize minimax game formulations to jointly optimize policies and reward discriminators. In this setting, the policy acts as a generator seeking to maximize expected preference, while a discriminator approximates the human preference function, regularized to prevent overfitting to spurious correlations in the static dataset. Such adversarial setups are particularly effective in handling intransitive or stochastic preferences, as they frame the learning process as finding a Nash equilibrium rather than maximizing a potentially misspecified scalar reward [103; 102]. Theoretical analysis suggests that these direct preference optimization methods can be reduced to robust reward-based RL, offering sample complexity guarantees comparable to traditional approaches while bypassing the instability of separate reward modeling [35]. Furthermore, incorporating uncertainty quantification directly into the reward inference process is essential for robustness. Methods that penalize reward estimates for state-action pairs exhibiting high variance across an ensemble of reward models prevent the policy from exploiting ambiguous regions of the preference landscape, a technique shown to mitigate overoptimization in offline settings [130].

Despite these advancements, significant trade-offs remain between the expressiveness of the reward model and the conservatism required for safe policy extraction. While direct methods like Contrastive Preference Learning eliminate the need for explicit reward functions by optimizing policies via contrastive losses on trajectory segments, they may struggle with non-Markovian dependencies unless augmented with transformer-based architectures [100; 44]. Conversely, model-based approaches that learn dynamics alongside rewards offer better sample efficiency through synthetic rollouts but require rigorous uncertainty penalization to avoid compounding model errors [65]. Future directions point towards hybrid architectures that integrate latent variable models to capture pluralistic human preferences within a single joint optimization framework, allowing robots to adapt to diverse user intents from heterogeneous offline datasets without catastrophic forgetting [96]. However, as static policies derived from these joint optimizations inevitably face evolving user intents and non-stationary environments, the focus must shift from extracting a single aligned policy to developing mechanisms for continual adaptation. This necessity paves the way for integrating sequential preference streams to address dynamic operational contexts, balancing the plasticity for new feedback with the stability to prevent catastrophic forgetting.

5.3 Continual Adaptation and Evolutionary Preference Alignment

The integration of sequential preference streams into offline robotic controllers necessitates frameworks that balance plasticity for evolving user intent with stability to prevent catastrophic forgetting. As operational environments shift and human operators refine their criteria, static policies derived from historical data become suboptimal, demanding continual adaptation mechanisms that leverage new feedback without discarding previously acquired competencies. A primary strategy involves meta-learning architectures that encode a prior over preference distributions, enabling rapid adaptation to individual user styles or changing objectives through context-based inference from sparse queries. For instance, [127] demonstrates that pre-training preference models on diverse task data allows robots to adapt to new tasks with significantly reduced online feedback, effectively mitigating the cold-start problem inherent in continual settings. Similarly, [33] employs latent variable formulations to infer user-specific preferences, allowing a single model to accommodate divergent human values without retraining from scratch, thereby preserving the capacity to switch between personalized behaviors dynamically.

To address the temporal non-stationarity of reward landscapes, recent approaches incorporate regularization techniques that constrain policy updates to critical parameters while allowing flexibility in others. The challenge of cross-domain continual learning is notably addressed by [131], which proposes a dual-module architecture simulating intraspecific competition. By utilizing fast and slow chasing modules with different optimization speeds, this method facilitates rapid adaptation to new preferences while the slow module retains historical knowledge, effectively mitigating forgetting in cross-domain scenarios. This evolutionary perspective is further extended by [103], which frames preference aggregation as a zero-sum game to robustly handle intransitive and stochastic preferences that often arise when aggregating feedback over time from multiple users or shifting criteria.

Furthermore, the composition of complex behaviors over long horizons requires mechanisms that can dynamically re-chain skills based on incoming preference signals. [132] offers insights into autonomous skill discovery, where agents self-generate and order goals, a principle adaptable for continually stitching skills guided by evolving human preferences. In multi-objective settings, where user trade-offs may drift, [133] provides a foundation for learning a universal policy representation that can instantly execute optimal behaviors for any given preference vector, supporting seamless adaptation to changing task definitions. However, reliance on implicit reward models, as seen in direct preference optimization, poses risks in continual settings due to limited generalization capabilities under distribution shift, as noted in [105]. Consequently, hybrid approaches that distill robust reward models or employ replay buffers of critical preference examples, as suggested in the context of robust preference optimization [134], are essential for maintaining alignment integrity. Future research must therefore focus on developing theoretically grounded bounds for simple regret in non-stationary preference environments and creating memory-efficient architectures that can scale lifelong preference alignment without compromising the safety guarantees established by the initial offline dataset.

5.4 Cross-Task Transfer and Zero-Shot Preference Generalization

Building on the necessity for continual adaptation discussed previously, the transfer of preference-aligned policies across disparate robotic tasks and embodiments represents a critical frontier in scaling offline preference-based reinforcement learning (PbRL). While existing methods effectively align policies within a single task domain, the prohibitive cost of collecting dense human preference

annotations for every new environment necessitates frameworks capable of zero-shot or few-shot generalization. This subsection synthesizes approaches that decouple the structural understanding of dynamics from the semantic interpretation of human intent, leveraging representation learning and optimal transport to bridge domain gaps.

A primary strategy for cross-task transfer involves aligning trajectory distributions between source and target domains using structural metrics that are invariant to state-space differences. Techniques utilizing offline datasets to craft preference queries and learn reward functions enable agents to perform novel tasks not explicitly shown in the data, effectively inferring intent based on structural similarities in state-action spaces [43]. This approach allows agents to generalize human intent without requiring explicit correspondence between state features, addressing the challenge where reward functions learned in one domain fail to transfer due to causal confusion or spurious correlations [3]. Furthermore, representation transfer frameworks enhance this capability by learning shared latent representations from multi-task offline data while explicitly penalizing representation uncertainty in the target task. By quantifying pointwise uncertainty in learnt representations, algorithms can mitigate the effects of incomplete coverage, enabling the transfer of reward structures even when the target task exhibits significant distributional shift from source domains [135; 136].

Beyond structural alignment, recent advancements exploit the multi-objective nature of human preferences to achieve generalization across unseen trade-offs, extending the cross-domain strategies mentioned earlier. Methods employing deep reinforcement learning can compute convex coverage sets containing potential optimal solutions for unknown objective importances, extending the Pareto front beyond the narrow distribution of preferences present in the original offline dataset [137; 133]. These approaches treat preferences not as static scalars but as directional vectors in a reward space, allowing for arithmetic control over policy behavior in novel contexts [128]. Similarly, meta-learning architectures encode a prior over user preference distributions, facilitating rapid adaptation to individual user styles or changing criteria via context-based inference from few-shot queries [127; 138]. This is particularly effective when combined with skill-based decompositions, where incoming preference signals dynamically re-weight modular latent skills learned from heterogeneous offline data, facilitating the composition of novel behaviors while retaining the stability of underlying primitives [139; 89].

Despite these advances, significant challenges remain regarding the robustness of transfer under hidden context and pluralistic values. Standard preference learning often implicitly aggregates diverse human values via Borda count, which can produce counter-intuitive results when transferring across populations with divergent cultural or contextual norms [32]. Future research must therefore focus on developing causal preference models that disentangle invariant user intent from domain-specific confounders, ensuring that transferred policies remain aligned with human values even in out-of-distribution settings [140]. Additionally, integrating foundation models for semantic reward specification offers a promising avenue for zero-shot transfer, where high-level language descriptions of intent guide policy adaptation in physically distinct environments [115; 141]. Ultimately, achieving robust zero-shot preference generalization requires a unified theory that jointly optimizes for dynamics coverage, reward identifiability, and semantic consistency across the manifold of robotic tasks.

6 Evaluation Protocols, Benchmarking, and Real-World Deployment

6.1 Standardized Benchmarks and Dataset Curation for Offline Preference Learning

The evaluation of offline preference-based reinforcement learning (RL) hinges critically on the fidelity and diversity of benchmark datasets, which serve as the bedrock for validating algorithmic robustness against distributional shift and reward ambiguity. Historically, the field has relied on synthetic replay buffers generated by online agents, such as those in [142] and [143]. While these repositories offer controlled environments for studying extrapolation error, they often fail to capture the heterogeneity and sub-optimality inherent in real-world robotic data collection. Consequently, a paradigm shift has emerged toward benchmarks comprising human demonstrations and unstructured internet-scale video, exemplified by [41] and [59]. These datasets introduce complex multimodal distributions and “type mismatches” between visual observations and action spaces, presenting a more rigorous testbed for generalization than agent-generated logs.

However, the transition to human-centric data exposes a significant gap in preference annotation density. Unlike standard offline RL benchmarks where scalar rewards are implicitly defined by task success, preference-based learning requires explicit ordinal relations between trajectory segments. The [144] suite addresses this by simulating teachers with varying degrees of irrationality, yet it largely relies on simulated ground-truth rewards to generate preferences, potentially overlooking the nuances of genuine human cognitive bias. Recent efforts like [28] challenge the standard Bradley-Terry-Luce assumption that preferences derive solely from partial returns, proposing instead that human judgments are informed by regret or deviation from optimality. This theoretical insight underscores a limitation in current benchmarks: many fail to account for non-Markovian temporal dependencies or the “causal confusion” where agents exploit spurious correlations rather than true task objectives, as detailed in [3].

Furthermore, the quality of preference datasets is increasingly scrutinized through the lens of conflict and pluralism. Standard benchmarks often assume a monolithic reward function, ignoring the reality of diverse human values. The introduction of metrics like “Alignment Dimension Conflict” in [145] and frameworks for modeling heterogeneous preferences in [96] highlight the necessity for benchmarks that capture conflicting alignment objectives. Without such diversity, algorithms risk overfitting to specific annotator biases or failing to generalize across user populations. Additionally, the susceptibility of these datasets to poisoning attacks, as demonstrated in [146], necessitates benchmarks that evaluate robustness not just to noise, but to adversarial corruptions in preference labels.

Looking forward, the curation of standardized benchmarks must evolve to include dense, multi-annotator preference labels alongside raw trajectories to facilitate the development of robust reward models. The integration of datasets that explicitly model hidden contexts and divergent user intents, as suggested by [32], will be essential. Future benchmarks should also prioritize the evaluation of “offline hyperparameter selection” mechanisms, as noted in [147], to ensure that model validation can occur without unsafe online rollouts. By bridging the gap between synthetic simplicity and the messy reality of human feedback, the community can establish evaluation protocols that truly reflect the challenges of deploying aligned robotic systems in dynamic environments.

6.2 Off-Policy Evaluation and Safety Verification without Environment Interaction

Building upon the necessity for robust benchmark curation and multi-annotator preference datasets, the final barrier to safe deployment lies in rigorously validating these policies without physical in-

teraction. Off-policy evaluation (OPE) serves as this critical gatekeeper for safety-critical robotic systems, enabling the estimation of value, risk, and constraint satisfaction solely from static historical datasets. The fundamental challenge remains correcting for the distributional shift between the behavior policy that generated the data—often heterogeneous and sub-optimal as noted in previous discussions—and the target policy being evaluated. Traditional Importance Sampling (IS) estimators frequently fail in this regime due to variance that scales exponentially with the horizon [18]. To mitigate this “curse of horizon,” recent advancements have shifted toward marginalized importance sampling and doubly robust estimators, which combine model-based predictions with weight corrections to achieve unbiasedness with significantly lower variance [148; 149]. Specifically, methods like [149] formulate OPE as a minimax game to directly estimate density ratios without explicit knowledge of the behavior policy, offering asymptotic optimality in tabular settings and robustness in function approximation regimes.

Beyond point estimates of expected return, rigorous safety verification necessitates characterizing the full distribution of returns to assess worst-case scenarios, a requirement amplified by the conflicting alignment objectives and potential data poisoning risks identified in benchmark studies. Distributional OPE approaches, such as Fitted Likelihood Estimation (FLE), leverage generative models to estimate the return distribution under total variation or Wasserstein distance metrics, providing probabilistic guarantees on tail risks [150]. This is complemented by risk-averse criteria like Conditional Value at Risk (CVaR), which can be estimated via conservative distributional actors that penalize out-of-distribution quantiles, ensuring that selected policies satisfy safety constraints even under significant model uncertainty [52]. Furthermore, in partially observable environments where latent confounders bias standard estimators, proximal causal inference techniques construct pessimistic confidence regions to correct for unmeasured state dependencies, marking a significant theoretical leap for OPE in complex robotic perception tasks [151].

A pivotal practical advancement bridging dataset curation and physical deployment is the development of offline hyperparameter selection protocols that eliminate the need for online rollouts during model validation. By utilizing offline metrics such as Fitted Q-Evaluation (FQE) or classification-based proxies, practitioners can reliably rank candidate policies and tune hyperparameters strictly within the offline constraint, a capability demonstrated to be feasible even in high-dimensional visual domains [147; 152]. However, empirical studies caution that standard prediction errors in vision-based models do not always correlate with driving quality or control performance, necessitating the design of validation datasets and metrics specifically aligned with downstream task success [153]. Emerging trends also explore pairwise estimation frameworks, such as Δ -OPE, which reduce variance by exploiting the positive covariance between similar policies to estimate relative improvements rather than absolute values [154]. Future directions must address the integration of these estimators into distributionally robust frameworks that account for both dynamics uncertainty and reward ambiguity, ensuring that safety verification remains valid when deployed environments deviate from the nominal conditions captured in offline logs [91]. Having established these rigorous protocols for offline policy evaluation and hyperparameter selection, the subsequent challenge lies in bridging the reality gap to deploy these verified policies in physical environments.

6.3 Bridging the Reality Gap via Simulation-Augmented and Hybrid Workflows

Bridging the reality gap remains a pivotal challenge in deploying offline and preference-based robotic systems, necessitating workflows that synergize imperfect simulators with sparse real-world data. Purely data-driven offline approaches often struggle with limited coverage and the inability to generalize to out-of-distribution states, while purely simulation-based methods suffer from dynamics

mismatches that invalidate policies upon transfer. Hybrid workflows address these dual limitations by leveraging simulators for broad exploration and real-world offline traces for fidelity grounding. A central paradigm in this domain involves conservative offline simulation-based policy optimization, where agents are trained on synthetic data but regularized to prevent overfitting to simulator biases. For instance, [71] introduces a dynamics-aware evaluation scheme that adaptively penalizes Q-values for simulated state-action pairs exhibiting large dynamics gaps relative to real-world data, effectively creating a “trust region” for simulation usage. Similarly, [155] proposes reweighting simulated data using a discriminator trained to distinguish between real and simulated state distributions, ensuring that the policy prioritizes transitions consistent with physical reality.

Beyond reweighting, advanced techniques focus on inferring optimal domain randomization distributions directly from offline traces to construct high-fidelity randomized simulators. Rather than relying on hand-tuned parameter ranges, likelihood-based approaches estimate the posterior distribution of dynamics parameters that best explain the observed real-world trajectories. This allows for the generation of synthetic data that accurately reflects real-world uncertainty, facilitating zero-shot transfer. Complementing this, [156] offers a generative alternative by utilizing diffusion models to synthesize entire trajectories that balance the target policy’s objectives with the behavior distribution’s plausibility, effectively augmenting sparse datasets without requiring explicit dynamics modeling. This approach mitigates the compounding errors typical of autoregressive world models while providing diverse, on-policy synthetic experience.

Hybrid pre-training and fine-tuning pipelines further exemplify this synthesis, where large-scale simulated or internet video data initializes representations or value functions, followed by adaptation using limited real-world offline datasets. [94] demonstrates that pre-training value functions on unlabelled human videos yields representations significantly more conducive to downstream robotic tasks than training from scratch, effectively bypassing the “type mismatch” between video observations and robotic actions. Furthermore, [157] advocates for regularizing online planners with epistemic uncertainty estimates derived from offline pre-training, enabling few-shot adaptation to novel tasks while maintaining safety constraints. Despite these advancements, critical trade-offs persist; overly conservative regularization may stifle the beneficial generalization capabilities of simulators, while aggressive synthetic data augmentation risks introducing stealthy distributional shifts. Future directions point toward causal inference methods to disentangle invariant physical laws from domain-specific artifacts, allowing for more robust sim-to-real transfer even under significant morphological or environmental changes.

6.4 Human-in-the-Loop Assessment and Alignment Verification

Successfully navigating the deployment challenges of hybrid simulation-to-real workflows sets the stage for the final validation of robotic systems, where the focus shifts from technical fidelity to holistic assessments of human-robot alignment and preference satisfaction. This critical transition necessitates moving beyond scalar performance metrics, such as task success rates, which often fail to capture the nuance of human intent, particularly when preferences are latent, pluralistic, or evolving. Consequently, evaluation protocols must integrate adaptive interfaces that mitigate the cognitive burden on annotators while enhancing the fidelity of collected feedback. Advanced elicitation frameworks now employ feature highlighting and outlier detection to guide users toward informative comparisons, significantly reducing label noise and bias during the data collection phase [158]. Furthermore, the reliability of these assessments hinges on the system’s ability to handle inconsistent or conflicting human inputs, a common reality in crowd-sourced or multi-user environments. Robust evaluation metrics must therefore quantify not only the convergence to a

single reward function but also the policy’s resilience to preference corruption and its capacity to model hidden contexts arising from diverse annotator backgrounds [32; 159].

A critical challenge in this alignment verification is distinguishing between genuine preference satisfaction and reward hacking driven by spurious correlations. Empirical studies indicate that policies optimizing misidentified rewards often exhibit high predicted utility while failing catastrophically in out-of-distribution states, a phenomenon exacerbated by non-causal distractor features in visual domains [3]. To address this, modern assessment protocols incorporate causal disentanglement techniques and stress-test policies against environmental perturbations to ensure that learned behaviors generalize beyond the training distribution. Moreover, the static nature of many offline datasets necessitates dynamic evaluation strategies that can assess a robot’s adaptability to shifting user needs. Few-shot adaptation protocols, where policies are fine-tuned via minimal interaction, have emerged as a vital metric for measuring personalization capabilities, demonstrating that meta-learned priors can drastically reduce the query complexity required for effective alignment in novel tasks [127].

The assessment of personalized interaction styles further requires moving beyond aggregate performance scores to individualized satisfaction metrics. Approaches that model user preferences as directional vectors in a multi-objective space allow for the arithmetic customization of behaviors, enabling evaluators to verify if a system can seamlessly switch between conflicting objectives such as safety versus efficiency [128]. Additionally, the integration of diffusion-based planners conditioned on attribute strengths provides a mechanism for zero-shot behavior customization, offering a new frontier for evaluating how well robots can synthesize unseen trajectories that adhere to abstract human instructions [83]. Despite these advancements, significant gaps remain in standardizing longitudinal user studies that capture the temporal dynamics of trust and comfort. Future evaluation frameworks must therefore evolve to include continuous, interactive feedback loops where humans can correct behaviors in real-time, bridging the divide between static offline optimization and the fluid nature of physical human-robot interaction [160]. Ultimately, true alignment verification demands a symbiotic evaluation pipeline where algorithmic robustness is inextricably linked to qualitative human experience.

7 Conclusion

This subsection synthesizes the convergence of offline reinforcement learning (RL) and preference-based RL, marking a paradigm shift from reward-engineered online exploration to data-constrained, human-aligned autonomy. The field has matured through three critical advancements: robust policy optimization under distributional shift, sophisticated modeling of latent human intent, and the integration of these paradigms into unified frameworks. Algorithmically, the transition from naive behavior cloning to pessimistic value estimation has been pivotal. Methods such as Conservative Q-Learning and distributional approaches like [52] have established that penalizing out-of-distribution actions is essential for safety in static datasets. Concurrently, sequence modeling architectures have enabled “trajectory stitching,” allowing agents to compose optimal behaviors from suboptimal offline logs, thereby overcoming the limitations of strict imitation [11].

In parallel, preference modeling has evolved beyond the standard Bradley-Terry-Luce assumption of partial returns. Recent theoretical insights suggest that human preferences are better modeled via regret minimization, offering stronger identifiability guarantees for latent reward functions [28]. This has spurred the development of direct preference optimization (DPO) algorithms that bypass explicit reward modeling, reducing the compounding errors inherent in two-stage RLHF pipelines [100]. Furthermore, the recognition of hidden context and pluralistic human values has led to distri-

butional preference learning frameworks that account for heterogeneous annotator biases, moving beyond the assumption of a single universal reward [32].

Despite these strides, critical theoretical gaps remain. A primary challenge is the disconnect between global coverage requirements for theoretical optimality in offline contrastive methods and the partial coverage typical of real-world robotic datasets [30]. While pessimistic algorithms like DCPPO offer guarantees under single-policy coverage for dynamic discrete choice models, extending these bounds to general function approximation in high-dimensional continuous control remains an open problem [36]. Additionally, the susceptibility of learned reward models to causal confusion and spurious correlations poses a significant risk; optimizing misidentified rewards can drive policies into unsafe out-of-distribution regions even when training error is minimal [3].

Future research must address these fragilities by integrating causal inference to disentangle true task objectives from environmental artifacts. Emerging frontiers include the fusion of foundation models for semantic reward specification and the development of multi-party RLHF frameworks that explicitly optimize for social welfare functions rather than aggregating diverse preferences into a single scalar [161]. Moreover, the integration of synthetic preference generation and counterfactual data augmentation offers a pathway to enhance sample efficiency without costly human labeling [162; 73]. Ultimately, the path to universally aligned robotic systems requires rigorous off-policy evaluation protocols that verify safety and alignment prior to deployment, ensuring that the pursuit of performance does not compromise the robustness necessary for real-world interaction [42].

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